

Can these layouts run 24 h continuously without failing an actuator?

Answer: yes. Worst case, a joint holds **61 % of the load the datasheet already warrants the arm can carry continuously** — and that is with the full 7.5 kg rated payload attached. Unloaded, as the scenes are currently built, it is 38 %.

Generated by `workspace_torque.py` ; raw values in `waypoint_torques.csv` . Specs from the [UR7e technical specification](#) (April 2026 revision).

The number that matters

Failure limits are **per joint**. Peak sustained holding torque on the hardest-worked joint, over the 26 waypoints all three layouts can hold:

layout	worst joint	unloaded	+ 7.5 kg rated payload	% of 150 N·m
urgantry	shoulder_lift	35.0 N·m	70.3 N·m	47 %
urtable	shoulder_lift	41.2 N·m	91.5 N·m	61 %
urtable_45	elbow / lift	56.5 N·m	69.2 N·m	46 %

Wrists stay under 43 % of their smaller ± 28 N·m limit in every case (`urtable_45` is the only layout that loads a wrist meaningfully).

Adding the rated payload roughly doubles the peaks and **reverses the ranking**: unloaded, `urtable_45` works an actuator hardest; loaded, `urtable` does, because its vertical bases put everything on `shoulder_lift` .

Why this is safe, from the datasheet alone

UR does not publish joint torque ratings — the UR7e sheet has none, and its only torque figures are the tool-flange F/T *sensor* range (± 10 N·m). But the payload rating implies the answer:

UR7e: 7.5 kg payload, 850 mm reach, any mounting orientation, 0-50 °C ambient.

A payload rating is a continuous guarantee. Holding 7.5 kg at full reach requires

- payload: $7.5 \text{ kg} \times 9.81 \times 0.85 \text{ m} = \mathbf{62.5 \text{ N}\cdot\text{m}}$
- arm self-weight: $20.6 \text{ kg} \times 9.81 \times \sim 0.35 \text{ m} \approx \mathbf{71 \text{ N}\cdot\text{m}}$
- $\approx \mathbf{133 \text{ N}\cdot\text{m}}$ sustained at the shoulder

That is the load UR warrants the joint can hold indefinitely, in any orientation, at up to 50 °C ambient. **Our worst measured case is 91.5 N·m — 69 % of it, with the same payload.** No layout asks the arm for anything it is not already specified to do continuously.

This also corroborates the model's 150 N·m `forcerange`: the arm's own spec implies ~133 N·m of worst-case static demand, so a 150 N·m limit is the right order of magnitude (89 % of it), which is what a real design would look like.

Practical note

e-Series joints have mechanical holding brakes that engage when the arm is stopped or powered down — position held with no motor current and no actuator load. Everything above assumes 24 h of *active servo-holding*, the worst case. Engaging the brakes for long static holds removes the question entirely.

Method

- Both arms driven to each waypoint by IK, tool down, each offset from the waypoint by 0.35 m along the base-to-base axis; loose props removed from the scene.
- `implicitfast` integrator — with the scenes' default Euler, the stiff position servos (kp 2000 / kv 400) chatter at the timestep frequency and no pose reads as settled.
- Physics runs ≥ 3 s then until $\max |\dot{q}| < 1e-3$ rad/s (cap 8 s); torques averaged over the following 0.25 s. A waypoint counts only if both arms solved IK and settled within 20 mm — 33 / 33 / 36 of 45 per layout, 26 in common.
- Payload figures add a 7.5 kg point mass at the TCP analytically ($\tau = -J^T \cdot F$) on top of the measured holding torque, rather than re-simulating with a loaded gripper.

Caveats

- **Static holding only.** Motion adds inertial torque this does not capture. The 133 N·m datasheet-implied figure is likewise static.
- **150 / 28 N·m is the MuJoCo model's `forcerange`**, inherited from the Menagerie UR5e, not a published UR7e rating. The datasheet-derived argument above does not depend on it.
- **The arm CoM distance (~0.35 m) is an estimate**, not a published figure; the 133 N·m is therefore approximate. The payload term (62.5 N·m) is exact from datasheet values.
- **26 of 45 waypoints.** Each layout's own-set figures cover different poses.
- Simulation: no friction, gearbox losses, or brake behaviour modelled.
- Earlier versions of this report ranked the layouts differently. Two measurement faults (integrator chatter dropping two-thirds of waypoints; a settle threshold loose enough to sample torques mid-creep) were found and fixed, and duty-cycle conclusions were originally drawn from mean rather than peak torques. The numbers above are from the corrected run and reproduce to 0.1 % under an independently written check.